



Development of a Scale-Down Model of hydrodynamic stress to study the performance of an industrial CHO cell line under simulated production scale bioreactor conditions

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ABSTRACT

The objective of this study was to develop a Scale-Down Model of a hydrodynamic stress present in large scale production bioreactors to investigate the performance of CHO cells under simulated production bioreactor conditions. Various levels of hydrodynamic stress were generated in 2 L bioreactors mimicking those present in different locations of a large scale stirred tank bioreactor.

In general, it was observed that tested cells are highly robust against the effect of hydrodynamic stress. However, at elevated hydrodynamic stress equivalent to an average energy dissipation rate, $\langle \varepsilon \rangle$, equal to 0.4 W/kg, the specific monoclonal antibody productivity, q_{mAb} , decreased by 25% compared to the cultivation conditions corresponding to $\langle \varepsilon \rangle$ equal to 0.01 W/kg. Even stronger decrease of q_{mAb} , in the order of 30%, was observed when $\langle \varepsilon \rangle$ was periodically oscillating between 0.01 and 0.4 W/kg to simulate the repeated passage of cells through the highly turbulent impeller discharge zone of a production scale bioreactor. Despite this effect, no changes in metabolite consumption or byproduct formation were observed. Furthermore, considering the experimental error product quality was independent of the applied $\langle \varepsilon \rangle$. To achieve a molecular insight into the observed drop of cellular productivity, a transcriptome analysis using mRNA microarrays was performed. It was found that transcripts related to DNA damage and repair mechanisms were upregulated when high $\langle \varepsilon \rangle$ was applied for cultivation.

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1. Introduction

Chinese Hamster Ovary (CHO) derived cell lines are the main industry workhorses for therapeutic protein production (Jayapal et al., 2007). The transfection methods typically applied today are a significant source of variation between CHO cell lines and also between different clones of the same cell line, making predictions of the performance of new cell lines under production conditions difficult. A further complication results from the fact that initial clone

screening is commonly done in small volume orbitally shaken systems, where both stirring and sparging are absent. On the contrary, production stirred tank reactors (STRs) with sparging are the most commonly used cultivation vessel type in industrial cell culture technology (Ma et al., 2006). While both stirring and sparging are necessary at large scale to supply the cells in culture with oxygen and nutrients for growth and production (Ozturk, 1996), they are also a source of hydrodynamic stress that could cause cell damage or affect cell's performance (Chisti, 2000). Therefore, the key question how to select the right cell for production environment already at the process development stage still remains unanswered.

To quantify hydrodynamic stress it is common to use the energy dissipation rate, ε , a scalar value that describes the irreversible conversion of kinetic energy to heat (Mollet et al., 2004; Nienow, 1998). The application of ε [W/kg or m²/s³] or corresponding power input per unit volume P/V [W/m³] as scaling criteria have been applied for scale-up and scale-down of impeller agitation rates in industrial cell culture processes (Li et al., 2006; Yang et al., 2007).

It is well documented that distribution of energy dissipation rate in stirred tank bioreactors is rather heterogeneous (Mollet et al., 2004; Nienow, 1998). Local ε values strongly depend on the

Abbreviations: ANOVA, analysis of variance; bG0, nongalactosylated glycan structure; bG1, monogalactosylated glycan structure; bG2, bigalactosylated glycan structure; HighM, high-mannose isoforms; CHO, Chinese Hamster Ovary; VCD, viable cell density [cells/mL].

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List of symbols

D_{imp}	impeller diameter [m]
Fl	flow number
N	agitation rate [1/s]
Q	impeller pumping capacity [m ³ /s]
mAB	specific monoclonal antibody production rate [pg/cell d]
q_{Glucose}	specific glucose consumption rate [mmol/cell d]
$q_{\text{Glutamine}}$	specific glutamine consumption rate [mmol/cell d]
q_{Lactate}	specific lactate production rate [mmol/cell d]
q_{Ammonia}	specific ammonia production rate [mmol/cell d]
t_c	circulation time [s]
t_M	overall mixing time [s]

Greek symbols

ε	energy dissipation rate [W/kg]
$\varepsilon_{\text{prod}}$	local ε in production bioreactor [W/kg]
$\langle \varepsilon \rangle$	average energy dissipation rate [W/kg]
$\langle \varepsilon \rangle_{\text{lab}}$	average energy dissipation rate in laboratory scale [W/kg]
$\langle \varepsilon \rangle_{\text{SDM}}$	Scale-Down Model of oscillating energy dissipation rate [W/kg]
$\varepsilon / \langle \varepsilon \rangle$	ratio of local and average energy dissipation rate

position inside the vessel. Moreover, this heterogeneity increases with stirred tank scale (Ducoste and Clark, 1998; Podgorska and Badyga, 2001; Wernersson and Tragardh, 1999). Several techniques have been established to quantify the heterogeneity of turbulence in a stirred vessel in terms of the ratio between local and averaged energy dissipation rates, $\varepsilon / \langle \varepsilon \rangle$. These include direct measurements, such as particle image velocimetry (PIV) (Gabriele et al., 2009; Zhu et al., 2009) and Laser Doppler anemometry (LDA) (Ducci and Yianneskis, 2005; Geisler et al., 1994; Zhou and Kresta, 1996b) and indirect ones, based on some shear sensitive model systems (Kresta and Wood, 1993; Podgorska and Badyga, 2001). For a single Rushton impeller, the largest values of $\varepsilon / \langle \varepsilon \rangle$ were found in the impeller discharge stream in the range of 20–60 (Baldi and Yianneskis, 2004; Ducci and Yianneskis, 2005; Escudie and Line, 2003; Sharp and Adrian, 2001; Zhou and Kresta, 1996b). In the case of pitched blade turbines, slightly lower values were found covering a range from 8 up to 40 (Delafosse et al., 2011; Gabriele et al., 2009; Zhou and Kresta, 1996b, 1998).

Beside agitation, sparging causes hydrodynamic stress that has been shown to be detrimental to cell cultures (Hu et al., 2011; Ma et al., 2006). High local ε occurs during bubble coalescence, breakup and burst (Boulton-Stone and Blake, 1993; Garcia-Briones et al., 1994). The dissipated energy during bubble disruption depends on the size of the bubble, being largest for smallest bubbles (Boulton-Stone and Blake, 1993; Garcia-Briones et al., 1994). This indicates that in large scale bioreactors, due to the large size of bubbles in the order of centimeters (Chisti, 2000), cell damage induced by bubble burst is substantially smaller than in small scale bioreactors, where bubble sizes are in the millimeter range.

Since sparging cannot be avoided in cell culture processes, especially in large scale (Henzler and Kauling, 1993; Ozturk, 1996), shear protective agents like serum and Pluronic F-68 were introduced and investigated thoroughly (Clincke et al., 2011; Ma et al., 2004; Michaels et al., 1991, 1995). Even though their protective mechanism is not yet completely understood, these substances effectively protect cells against shear forces and are therefore widely used in industrial cell culture (Gigout et al., 2008). The use of Pluronic F-68 is preferred over serum due to safety concerns associated with the animal origin of the serum and its cost. As shown by Ma et al. (2004)

at Pluronic F-68 concentrations of 1 g L⁻¹ or higher, the amount of cells attaching to rising bubbles is substantially reduced and therefore the impact of hydrodynamic stress generated during bubble rise and disruption is minimized.

Reported effects of shear forces on animal cells include direct lethal events leading to necrosis, lethal cellular responses caused by apoptosis induction and sub-lethal responses, e.g. changes of metabolism (Ma et al., 2006). The shear stress thresholds depend on the evaluated cell lines and the determination method, i.e. the flow conditions in the applied shear stress devices (Augenstein et al., 1971; Gregoriades et al., 2000; Ma et al., 2002; McQueen and Bailey, 1989; Mollet et al., 2007; Tanzeglock et al., 2009). Necrosis was observed in suspended CHO cells only after exposure to ε values of 10³–10⁵ W/kg (Godoy-Silva et al., 2009b; Ma et al., 2002; Mollet et al., 2007; Tanzeglock et al., 2009). In contrast, apoptosis is a cellular response which can be triggered by hydrodynamic stress and irreversibly leads to cell death (Al Rubeai et al., 1995; Fussenegger and Bailey, 1998). As demonstrated by Mollet et al. (2007) for CHO cells, apoptosis can be triggered already at ε values in the range of 10² W/kg. Furthermore, as shown by Godoy-Silva et al. (2009a), repeated exposure of an industrial CHO cell line to ε of 60 W/kg caused a sub-lethal effect, although the lethal ε value for this cell line was much larger, i.e. 6.4×10^3 W/kg. In particular, a shift in the antibody glycosylation pattern was observed, which is a critical quality attribute for therapeutic antibodies (Abu Absi et al., 2010; Berridge et al., 2009). Due to the sub-lethal nature of such responses, it is reasonable to expect that gene expression regulation may also be involved. Accordingly, mRNA microarrays could be potentially used to identify stress response mechanisms and gene-phenotype relationships (Griffin et al., 2007).

The aim of this work is to establish a Scale-Down Model of production bioreactors to test the performance and robustness of a new CHO cell line at various levels of hydrodynamic stress comparable to those found in a production scale bioreactor. An established CHO monoclonal antibody production fed-batch process was analyzed to derive the Scale-Down Model parameters for the new cell line (Palomares et al., 2010). A 2 L bioreactor was used as a single vessel Scale-Down Model operated at different agitation rates. Agitation rates changing periodically in time were applied to mimic the repeated exposure of cells to different local values of ε present in different locations inside the large scale bioreactor. The local ε values were derived from theoretical analysis and literature values. Analysis of cell growth and metabolism combined with measurements of product quality as well as gene expression were used to characterize the cellular responses.

2. Materials and methods

2.1. Cell line and process

In this work we use a Chinese Hamster Ovary (CHO) derived cell line that was adapted to suspension growth in serum-free media. Cells were stably transfected using standard methods (Jayapal et al., 2007) to produce a therapeutic human monoclonal antibody.

Cells were thawed from cell bank vials and expanded in shake flasks in chemically defined expansion medium. The expansion consisted of at least 4 subcultivation steps. Bioreactors were inoculated to a cell density of $0.5 \pm 0.1 \times 10^6$ cells mL⁻¹ by adding the appropriate amount of expansion culture directly to the chemically defined production medium.

2.2. Bioreactor setup

2 L glass bioreactors (Applikon Biotechnology B.V., Schiedam, NL), equipped with DO, pH and temperature probes as well as two

Table 1
Hydrodynamic experimental conditions in the 2 L scale-down studies.

$\langle \varepsilon \rangle_{\text{lab}}$ [W/kg]	2 L agitation rate [min^{-1}]	$\varepsilon_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ ratio	Simulated large scale bioreactor compartment	No. of cultivations
0.001	74	0.1	Low turbulence zone	a
0.01	155	1	Bulk zone	3
0.06	300	6	(Standard 2 L agitation rate)	7
0.4	566	40	High turbulence zone	2
0.01–0.4	155–566	1–40	Scale-Down Model	3
1.015	776	100	(Corresponding tip speed)	a

^aCultivations at 0.001 and 1.015 W/kg were excluded from further investigations due to cell aggregation and foaming issues, respectively (see Section 3).

elephant ear impellers with a diameter of 45 mm, installed 70 mm apart from each other, were used in this study. Baffles were not present, but the multiple pipes and probes were considered sufficient to provide full reinforcement. A perforated tube sparger was used for aeration. For DO control, a combination of constant air flow at a rate of 1.2 vvh and oxygen sparging on demand was used. The pH was controlled using CO₂ sparging combined with NaOH addition. If required, an Antifoam C solution (Dow Corning Co., Wiesbaden, Germany) was added to the liquid surface to reduce foaming. All online and offline process values and measurements were fed into the Process Information Management System (PIMS) Lucullus (SecureCell, Schlieren, CH).

To quantify the amount of kinetic energy introduced into the bioreactors, the net power consumption was determined as the product of current and voltage of the stirrer engine measured for filled and empty reactor with standard setup at different agitation rates (Ascanio et al., 2004). Obtained power was used to evaluate vessel averaged energy dissipation rates of the small scale bioreactors, $\langle \varepsilon \rangle_{\text{lab}}$, specific power input P/V , as well as the corresponding Power number. Thus, in order to investigate the effect of different values of local energy dissipation rates $\varepsilon_{\text{prod}}$, present at various locations within a production bioreactor, experiments were performed in 2 L bioreactors by adjusting the agitation rate so that $\langle \varepsilon \rangle_{\text{lab}} = \varepsilon_{\text{prod}}$. The adopted hydrodynamic experimental conditions listed in Table 1 were selected to mimic low turbulence zone, average liquid bulk and impeller discharge stream in our production bioreactors (see Section 3.1). Experimental cultivations were always compared to the laboratory specific standard cultivation conditions. As a somewhat extreme condition, tip speed of production scale was applied, which for this study was about 2 m/s. Finally, to mimic repeated cell exposure to different local hydrodynamic conditions in a 13.5 m³ large scale production bioreactor, i.e. intense turbulence close to the stirrer and average turbulence in the bulk, cells were periodically exposed to high agitation rate over a short period of time followed by a longer period of lower agitation rate. A short Lucullus step sequence was programmed to adjust the agitation rates in the desired frequency.

2.3. Offline measurements

Cell densities and viability were assessed by an automated cell counting device with the trypan blue staining method (ViCell, Beckman Coulter, Nyon, CH). Offline pH, dissolved O₂ and CO₂, nutrients (glucose, glutamine) and basic metabolite concentrations (lactate, glutamic acid, ammonia) were measured with Bioprofiles 400 and Flex (Nova Biomedical, Waltham, USA). MAb concentration and amino acid concentrations were determined by standard HPLC methods. The product quality was analyzed with Size Exclusion Chromatography, Cation Exchange Chromatography and glycosylation pattern analysis (Berridge et al., 2009; Rudd et al., 1997). Metabolic rates were calculated as the ratio of the substrate concentration difference Δc_i , corrected for feed addition and sampling volume, and the average of the viable cell density $\langle \text{VCD} \rangle$ in the

period of two consecutive measurements Δt defined as (Adams et al., 2007):

$$q_i = \frac{\Delta c_i}{\langle \text{VCD} \rangle \Delta t} [\text{pmol cell}^{-1} \text{ d}^{-1}] \quad (1)$$

2.4. Bioinformatics workflow

CHO-specific mRNA microarrays (Affymetrix, Santa Clara, USA) including probes for all known polyadenylated transcripts of CHO cell lines were applied. Samples were taken at predefined culture stages during the process, i.e. exponential growth phase, peak viable cell density and late production phase. For each sample, a culture volume containing 10⁷ cells was transferred to certified RNase-free 15 mL Falcon tubes, centrifuged, and the cell pellets were shock frozen on dry ice immediately. After microarray processing according to Affymetrix standard protocol, data were analyzed with Genedata Expressionist (Genedata, Basel, Switzerland), performing an ANOVA test over the hydrodynamic stress levels studied. The resulting gene list of differentially expressed genes was evaluated using the Metacore database (GeneGo, St. Joseph, USA) to connect the genes to their functions.

3. Results and discussion

3.1. Scale-Down Model characterization

As an example of the characterization of bioreactor hydrodynamics, mixing times measured using phenolphthalein color alteration (Zlokarnik, 2005) are presented in Figure SI 1 as a function of the average energy dissipation rate, $\langle \varepsilon \rangle_{\text{lab}}$. It is seen that even for $\langle \varepsilon \rangle_{\text{lab}}$ values around 0.01 W/kg, the measured mixing times were below 10 seconds, indicating that no inhomogeneities, apart from those related to the hydrodynamic stress, were present in the system.

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The measured Power number, P_o , for the used 2 L bioreactors operated in turbulent regime ($Re > 10^4$), equipped with two elephant ear impellers was equal to 4.8. According to Nienow (1998), as the distance between the impellers was about 1.6 impeller diameters, the Power number of a single impeller can be estimated as 1/2 of that measured for the dual impeller configuration, thus resulting in a value of 2.4. This value is in good agreement with the value of 2.1 measured for similar impellers by Zhu et al. (2009). More results on the characterization of the hydrodynamics of these reactors are reported elsewhere (Sieck et al., 2009).

Based on these findings, $\langle \varepsilon \rangle_{\text{lab}}$ could be adjusted in 2 L bioreactors to mimic certain compartments of the production scale bioreactor using local values of the energy dissipation rate. In particular, the bulk zone was simulated with an $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ equal to 1 and the low turbulence zone with $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ equal to 0.1. To estimate the local ε at the impeller discharge zone experimental data measured in small scale stirred vessels of several liters

equipped with pitched blade impellers were initially considered. According to these measurements the ratio of $\varepsilon_{\max}$ and $\langle \varepsilon \rangle$ can reach values as large as 40 (Delafosse et al., 2011; Zhou and Kresta, 1996a) which, when considering the $\langle \varepsilon \rangle_{\text{prod}}$ equal to 0.01 W/kg, would result in $\varepsilon_{\max}$ equal to 0.4 W/kg. However, as reported by Wernersson and Tragardh (1999) for stirred vessels equipped with Rushton impellers with volumes covering a range from 0.6 to 22 m³, $\varepsilon_{\max}$ increases with increasing $\text{Re}_{\text{impeller}}$. To take this into consideration, $\varepsilon_{\max}$ in the production bioreactor was estimated using the isotropic turbulence approximation according to which $\varepsilon_{\max}$ is proportional to the $N^3 D_{\text{impeller}}^2$ (Kolmogorov, 1991; Pope, 2000; Richardson, 1922; Zhou and Kresta, 1996a,b). Taking into consideration impeller diameter and operating conditions in 2 L and 13.5 m³ bioreactors it was found that $\varepsilon_{\max}$ in the discharge zone of the large scale bioreactor can reach values approximately 10 times larger compared to the 2 L bioreactors, i.e. 4 W/kg. Since our target is later on to apply oscillating ε conditions during culture, to ensure that every cell will be exposed to ε values high enough for a screening of robust CHO cell lines, stirring speed was adjusted such that $\langle \varepsilon \rangle_{\text{lab}}$ was equal to 0.4 W/kg, resulting in $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ equal to 40. Additionally, as somewhat extreme condition an experiment was performed at an $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ ratio of 100 to test the feasibility of scaling down the agitation rate based on the impeller tip speed of the production bioreactor, which was about 2 m/s. In parallel to the abovementioned conditions the historically applied standard operating conditions for the 2 L bioreactors were tested for comparison

as well, corresponding to an $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ ratio of 6. Details of the agitation rates applied in 2 L bioreactors and the corresponding $\langle \varepsilon \rangle_{\text{lab}}$ values are summarized in Table 1. All experiments discussed so far were performed at constant $\langle \varepsilon \rangle_{\text{lab}}$, simulating the hypothetical situation that the cells would persist in that particular large scale compartment for the entire cultivation time.

On the other hand, in order to mimic the journey of a cell inside large scale bioreactors during cultivation, experiments with varying $\langle \varepsilon \rangle_{\text{lab}}$, oscillating between $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ ratios of 1 and 40 were performed (see Table 1). According to Palomares et al. (2010), the average period between two subsequent passages of a cell through the high intensity zone near the impeller should be equal to the simulated large scale bioreactor's circulation time t_c . According to Nienow (1998), t_c can be approximated as 1/5 of the overall mixing time t_M , which can be measured experimentally, e.g. by base addition (Xing et al., 2009). In the large scale bioreactor considered in this work, equipped with two elephant ear impellers of 0.7 m diameter, operated at 50 rpm at a working volume of 13.5 m³, the overall mixing time t_M was found to be 180 s (95% homogeneity criterion). Using this value, the corresponding circulation time t_c and therefore the cell exposure frequency to high ε was estimated to be equal to 36 s.

To match the duration of the cell exposure to that found in the large scale bioreactor, the residence time in the impeller discharge zone near the impeller was estimated from the ratio of impeller swept volume and flow rate $Q (= \text{Fl} N D_{\text{imp}}^3)$ generated by the impeller. The flow number Fl of an elephant ear impeller as measured by Simmons et al. (2007) equals 0.87, thus the corresponding residence time is approximately 2 s. By taking into account the time required for the step change of the agitation rate, which was around 1 s, the overall time of cell exposure to high stress values in the Scale-Down Model was set to 3 s.

It is worth noting that the level of dissolved oxygen in all experimental conditions, including those with oscillating agitation rates, was within a 10% band from the setpoint, indicating that the dissolved oxygen setpoint could be maintained in these conditions as well.

3.2. Cultivation results

The effect of chronic hydrodynamic stress on cell growth is shown in Fig. 1a, where viable cell density (VCD) is reported as a function of cultivation time. In order to compensate for batch-to-batch variation, VCDs were normalized with respect to the maximum VCD obtained at the standard operating conditions (see Table 1). It can be seen that the time evolution of VCD measured at rather different values of $\langle \varepsilon \rangle_{\text{lab}}$, covering a range from 0.01 to 1.015 W/kg, is very comparable. At 1.015 W/kg, corresponding to the tip speed of the production bioreactor and a $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}}$ ratio of 100, where vortex formation was observed resulting in bubble entrainment into the liquid and therefore substantial foam formation. Intense agitation causing bubble entrainment is known to reduce VCDs for in cell culture (Kunas and Papoutsakis, 1990; Michaels et al., 1996; Oh et al., 1989). However, Fig. 1a shows that this was not the case in this study, since the time evolution of the VCD is comparable with that measured under standard conditions. These findings do not agree with previously published results, even using industrial CHO cell lines, which showed a reduction in viable cell density with increasing the hydrodynamic stress above a certain point (Al Rubeai et al., 1995; Li et al., 2006; Tanzeglock et al., 2009). This underlines that the cell line investigated here is highly robust against agitation shear stress, even in comparison with other industrial cell lines.

In comparison to the experiments performed at constant $\langle \varepsilon \rangle_{\text{lab}}$, while following the same overall growth behavior, VCDs measured

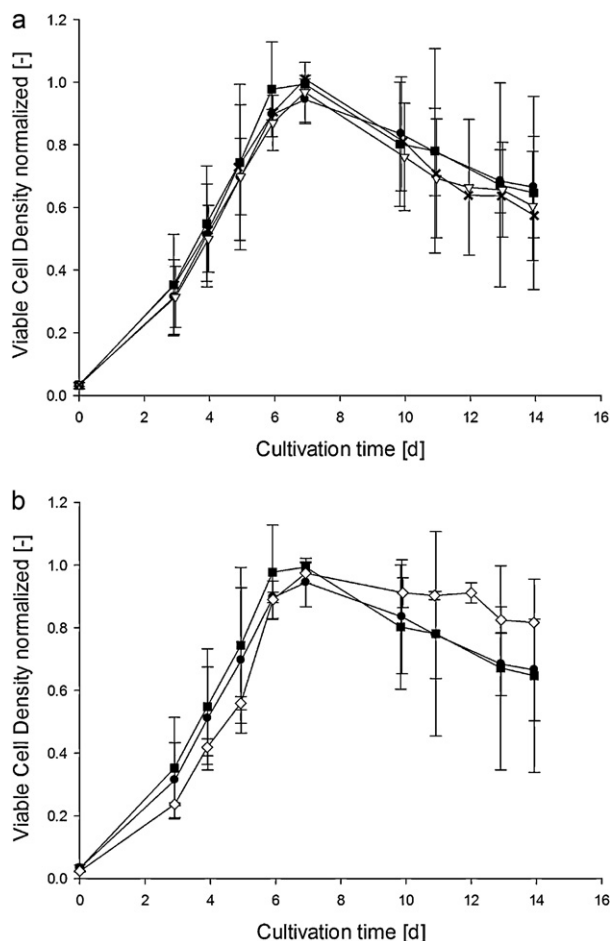


Fig. 1. Comparison of viable cell density measured during cultivation for various values of $\langle \varepsilon \rangle_{\text{lab}}$. (a) $\langle \varepsilon \rangle_{\text{lab}} = 0.01$ W/kg (solid circle), $\langle \varepsilon \rangle_{\text{lab}} = 0.06$ W/kg (open triangle), $\langle \varepsilon \rangle_{\text{lab}} = 0.4$ W/kg (solid square), and $\langle \varepsilon \rangle_{\text{lab}} = 1.015$ W/kg (cross). (b) $\langle \varepsilon \rangle_{\text{lab}} = 0.01$ W/kg (solid circle), $\langle \varepsilon \rangle_{\text{SDM}} = 0.01$ – 0.4 W/kg oscillating (open diamond), and $\langle \varepsilon \rangle_{\text{lab}} = 0.4$ W/kg (solid square).

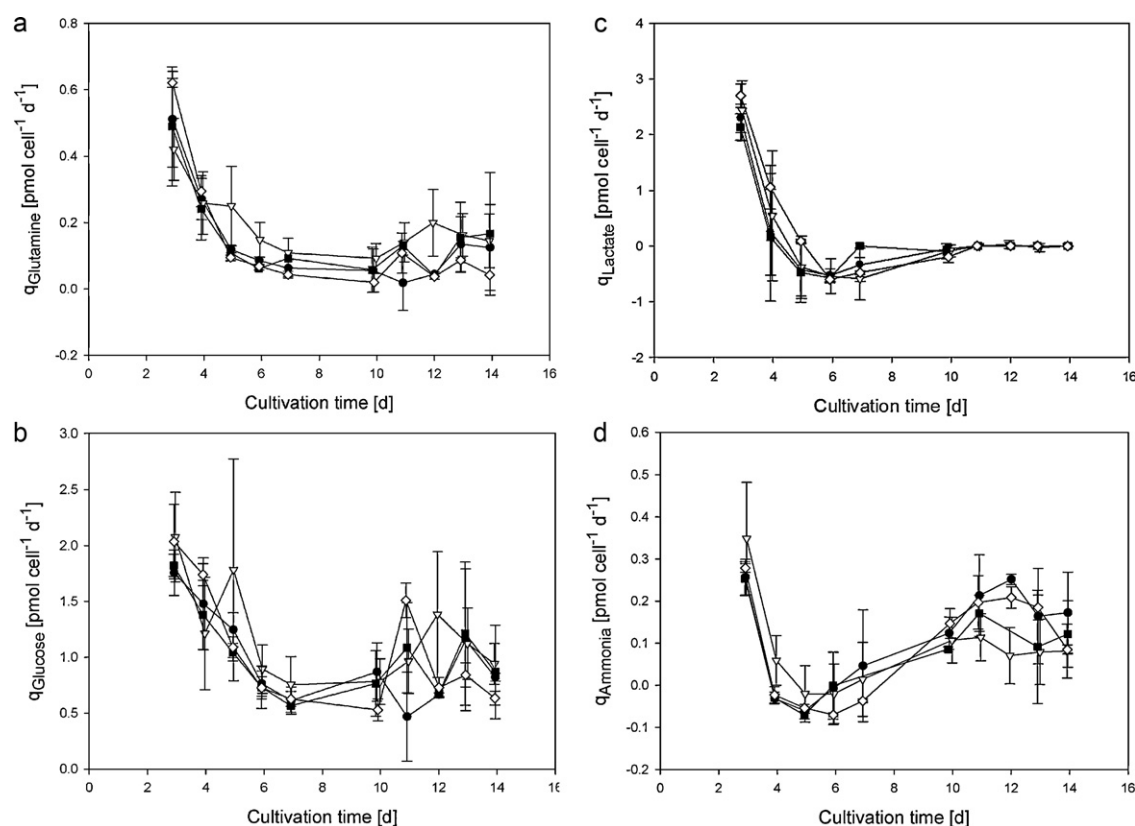


Fig. 2. Comparison of specific metabolic rates during the cultivation process. (a) Glutamine consumption, (b) glucose consumption, (c) lactate production, and (d) ammonia production. Symbols in all panels: (solid circle) $\langle \varepsilon \rangle_{\text{lab}} = 0.01 \text{ W/kg}$, (open triangle) $\langle \varepsilon \rangle_{\text{lab}} = 0.06 \text{ W/kg}$, (solid square) $\langle \varepsilon \rangle_{\text{lab}} = 0.4 \text{ W/kg}$, and (open diamond) $\langle \varepsilon \rangle_{\text{SDM}} = 0.01\text{--}0.4 \text{ W/kg}$ (oscillating).

in oscillating conditions revealed a lower growth rate during the first six culture days, while higher cell densities were achieved at the late stage of the cultivation, as shown in Fig. 1b.

It is worth noting, that the cultivation performed at a constant, very low $\langle \varepsilon \rangle_{\text{lab}}$ equal to 0.001 W/kg , simulating a low turbulence zone in the large scale bioreactor with $\langle \varepsilon \rangle_{\text{lab}} / \langle \varepsilon \rangle_{\text{prod}} = 0.1$, led to formation of cell aggregates, which indicates that a prolonged residence time in stagnant zones of large scale bioreactors may cause the same undesired phenomenon. For this, the VCD measurements for this cultivation were not comparable with the other cultivations. Due to the cell aggregation and the foaming issues, respectively, cultivations at 0.001 and 1.015 W/kg were excluded from further analysis in this work.

It is a common finding in the literature that stress in mammalian cell culture could result in an increased demand for energy metabolites (Godoy-Silva et al., 2009a; Keane et al., 2002; Ma et al., 2006), in this case glucose or glutamine. To evaluate the effect of hydrodynamic stress on cell metabolism, consumption and production of the main nutrients and metabolic products were investigated. Fig. 2 shows the specific glutamine and glucose consumption rates together with specific lactate and ammonia production rates as a function of cultivation time.

It is seen that both q_{Glucose} and $q_{\text{Glutamine}}$ decrease over the initial part of the cultivation while steady states are reached at later state of the cultivation at around 1 and $0.2 \text{ pmol cell}^{-1} \text{ d}^{-1}$, respectively (see Fig. 2a and b). No difference is observed in the specific consumption rates of glucose and glutamine for the various applied energy dissipation rates, irrespective of whether these are constant or oscillating. Limitations of other amino acids were also not observed in offline analysis for any of the cultivations (data not shown).

In Fig. 2c it is seen that lactate is being produced over the first 4 days of cultivation. After this point, a change in lactate metabolism results in lactate consumption until depletion around day 10. Ammonia production rate initially drops to a minimum at day 5–6, i.e. when maximum cell density is reached, and then increases during the decline phase of the culture, as shown in Fig. 2d.

The above results do not indicate that an alteration of cell metabolism, a limitation in energy metabolism or an inhibition by excessive byproduct formation, is caused by the tested energy dissipation rates, as all the metabolic rate time courses are substantially independent of the applied $\langle \varepsilon \rangle_{\text{lab}}$.

In Fig. 3, the specific monoclonal antibody production rate q_{mAb} for the different conditions is presented. Specific productivity shows a slight decrease at high hydrodynamic stress. For the constant exposure to the maximum $\langle \varepsilon \rangle_{\text{lab}}$ of 0.4 W/kg over the whole cultivation, a reduction of q_{mAb} of 25% was observed. Although in the oscillation experiments, this maximum value of $\langle \varepsilon \rangle_{\text{lab}}$ was present for only 10% of the cultivation time, this caused a comparably detrimental effect (reduction of q_{mAb} by 30%). This indicates that repeated changes in $\langle \varepsilon \rangle_{\text{lab}}$, which are a common phenomenon in stirred tank bioreactors, are themselves a stress effect for the considered cell line, resulting in specific productivity reduction.

A crucial point here is to determine, whether or not the quality of the produced antibody is affected by the level of hydrodynamic stress. The product quality was analyzed with Size Exclusion Chromatography, Cation Exchange Chromatography and glycosylation pattern analysis. It was found that in all cases the measured differences in product quality were very small and close to the limit of detection, indicating that the different values of the applied hydrodynamic stress do not affect product quality. However, it is worth

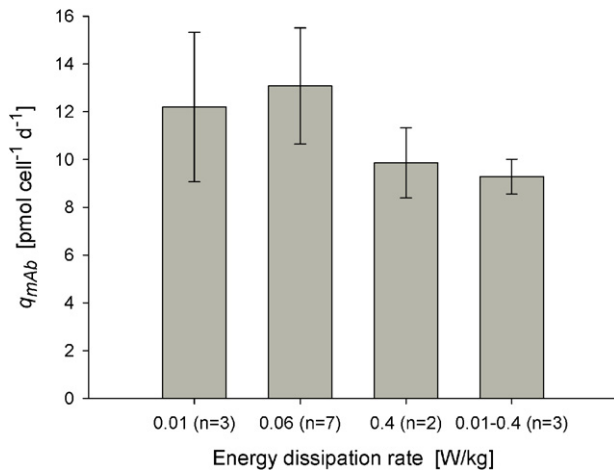


Fig. 3. Comparison of specific monoclonal antibody production rate q_{mAb} between day 6 and the end of the culture for various applied $\langle \epsilon \rangle_{lab}$.

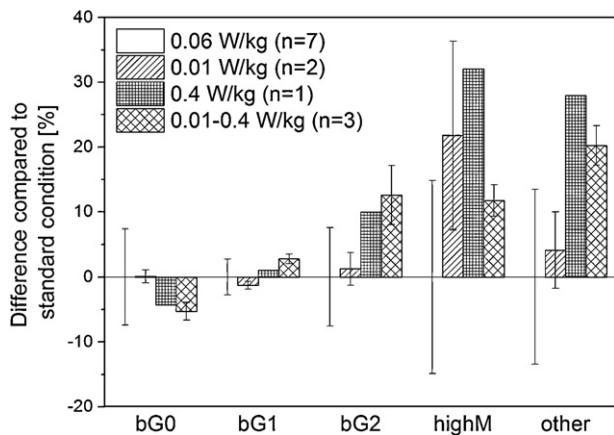


Fig. 4. Variation of glycosylation pattern obtained for different values of $\langle \epsilon \rangle_{lab}$. The bG0, bG1 and bG2 isoforms account for >90% of the total product.

noticing that the data obtained in this study exhibit a trend in the glycosylation pattern which is similar to the one described by Godoy-Silva et al. (2009a), who found an increase of galactosylated isoforms with increasing hydrodynamic stress. This is apparent in Fig. 4, where the glycopattern is shown for cultivations at various values of energy dissipation rate, $\langle \epsilon \rangle_{lab}$. The error bars represent one standard deviation, based on the repetition of standard conditions, i.e. $\langle \epsilon \rangle_{lab} = 0.06$ W/kg. The columns represent the difference between the specific experiment and the standard conditions. The isoform bG0 is the non-galactosylated structure, while bG1 and bG2 contain one and two terminal galactoses, respectively (Glycomics, 2012 #39). These three isoforms usually account for >90% of the total product. The high mannose (HighM) and other isoforms are present in relatively small amounts, close to the limit of detection, which explains the high variability for these columns in Fig. 4. A tendency toward higher galactosylation of the antibodies at high and oscillating energy dissipation rates can be observed. Galactose content is not considered a risk for drug safety, but it could affect the activity of therapeutic antibodies (Berridge et al., 2009). Thus, galactosylation should be closely monitored during scale-up.

In Fig. 5, reported levels of energy dissipation rate in bioreactors are listed as well as the lethal and sub-lethal responses of mammalian cell cultures to these hydrodynamic stresses. When comparing the threshold value of the average energy dissipation rate of 0.4 W/kg, beyond which, in our data, the reduction in mAb productivity takes place, with the point of the glycosylation pattern

Table 2

Metacore pathways differentially expressed under the considered experimental conditions.

Priority	Pathway maps (keywords)	p-Value
1	Cell cycle transition and termination of DNA replication	1.82E-05
2	DNA damage; DNA repair	1.44E-03
3	Apoptosis and survival	2.82E-03
4	Cell adhesion; ephrins signaling	3.23E-03
5	Immune response; histamine signaling	3.97E-03
6	DNA damage induced response	1.71E-02
7	DNA damage response	2.46E-02
8	CFTR expression	2.83E-02
9	DNA damage induced apoptosis	2.83E-02
10	G-protein signaling	3.20E-02

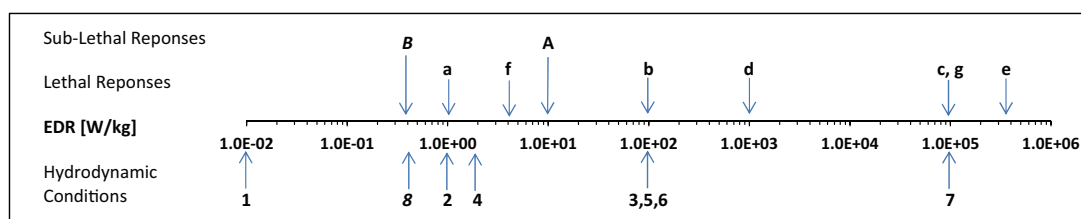
shift equal to 60 W/kg measured by Godoy-Silva et al. (2009a), it can be seen that our value is much lower, approximately by a factor of 150. It must be mentioned that the ϵ values in Godoy-Silva's study were generated by the passing of cells through a 'torture chamber', which is a laminar flow chamber and thus different to turbulent bioreactor flow. However, we believe that the major source of this difference is related to frequency and period of cell exposure to hydrodynamic stress, which were both higher in this work. In particular, the duration of the high stress exposure is equal to 3 s and it is repeating every 36 s while the sub-lethal effect observed by Godoy-Silva et al. (2009a) was triggered by an exposure frequency of 333 min and a duration of exposure was around 0.13 ms, estimated as the inverse of the applied shear rate. As this exposure time is relatively short and followed by a very long period in which cells could recover, it can be understood that our Scale-Down Model setup with both higher exposure time and frequency, could result in to lower threshold values for the shear stress.

3.3. Transcriptome analysis

To identify the mechanism underlying the loss in specific productivity, an analysis of the transcriptome was performed. Samples were taken at hydrodynamic stresses $\langle \epsilon \rangle_{lab}$ equal to 0.01, 0.06 and 0.4 W/kg (see Table 1), to which cells were exposed constantly during the whole cultivation. The obtained raw expression data were evaluated by Genedata Expressionist in a 2-Way ANOVA. In total, 57 transcripts were found to be 2-fold or higher up- or down-regulated when the cells were exposed to high hydrodynamic stress, which is a relatively small amount compared to the number of transcripts that change as a function of cultivation time. Also, statistical significance of the differential expression pattern was rather low ($p < 0.001$, $q = 0.3$).

The obtained transcriptome data were further evaluated using the Metacore database to identify functional cellular pathways affected by the hydrodynamic stress. It was found that 5 of the top 10 canonical pathways which were differentially regulated are related to DNA damage and repair mechanisms (Table 2). The p -values provided in Table 2 (top 10) range from 1.819×10^{-5} to 0.032 and give the statistical probability that the differential regulation of the identified pathways was a random event (false positive). The p -values were calculated by Metacore based on a statistical comparison of the set of 57 differentially expressed CHO transcripts against all mouse transcripts in each pathway available in the Metacore database. The Affymetrix microarray does not represent all these transcripts, thus the p -values should be considered overestimated (i.e. biological background correction was not applied).

The obtained data clearly indicate some effect of hydrodynamic stress on gene regulation. Since no mechanistic explanation for the reduced specific productivity can be derived from the microarray results, this can be regarded as a general stress response, not related to protein production specifically. However, the data suggest that,



Sub-Lethal Responses

Symbol	Energy Dissipation Rate [W/kg]	Mode of Cultivation	Shear Flow Type	Response	Reference
A	1.0E+01	Suspended	Extensional Flow	Shift in Glycopattern	Godoy-Silva et al. (2009)
B	4.0E-01	Suspended	Bioreactor	productivity loss, transcriptomic stress response	this work

Lethal Responses

Symbol	Energy Dissipation Rate [W/kg]	Mode of Cultivation	Shear Flow Type	Response	Reference
a	1.0E+00	Anchored	Extensional Flow	Necrosis	Gregoriades et al. (2000)
b	1.0E+02	Anchored	Extensional Flow	Apoptosis	Mollet et al. (2007)
c	1.0E+05	Suspended	Extensional Flow	Necrosis	Mollet et al. (2007)
d	1.0E+03	Suspended	Extensional Flow	Necrosis	Godoy-Silva et al. (2009)
e	3.6E+05	Suspended	Extensional Flow	Necrosis	Tanzeglock et al. (2009)
f	4.0E+00	Suspended	Plain Shear Flow	Necrosis	Tanzeglock et al. (2009)
g	1.0E+05	Suspended	Extensional Flow	Necrosis	Ma et al. (2002)

Hydrodynamic Conditions

Symbol	Energy Dissipation Rate [W/kg]	Description	Reference
1	1.0E-02	Volume average energy dissipation rate in typical large scale cell culture bioreactors	Varley and Birch (1999)
2	1.0E+00	Volume average energy dissipation rate in a 10L mixing vessel (Rushton Turbine, 700 rpm)	Zhou and Kresta (1996b)
3	1.0E+02	Maximum local energy dissipation rate in a 10L mixing vessel (Rushton Turbine, 700 rpm)	Zhou and Kresta (1996b)
4	2.0E+00	Volume average energy dissipation rate in a 22000L bioreactor (Rushton Turbine, 140 rpm)	Wernersson and Tragardh (1999)
5	1.0E+02	Maximum local energy dissipation rate in a 22000L bioreactor (Rushton Turbine, 140 rpm)	Wernersson and Tragardh (1999)
6	1.0E+02	Bubble rupture (pure water, bubble diameter 6.32 mm)	Garcia-Briones et al. (1994)
7	1.0E+05	Bubble rupture (pure water, bubble diameter 1.7 mm)	Boulton-Stone and Blake (1993) Garcia-Briones et al. (1994)
8	4.0E-01	Maximum local energy dissipation rate in 13500L bioreactor ('Elephant Ear' Impellers, 50 rpm)	this work

Fig. 5. Comparison of energy dissipation rate for transcriptomic shear stress response obtained in this work with thresholds for lethal and sub-lethal responses obtained by other authors. For completeness, data of typical energy dissipation rate present in stirred tank bioreactors are also reported including the corresponding references.

for the studied cell line, DNA damage may be part of the mechanism affecting cells at high ε . It should be mentioned that the Metacore database is not focused on CHO cells, thus the response pattern would be more characteristic if a different database would be used (Hammond et al., 2011), which was not possible in this case. To improve statistical significance and informative value of mRNA microarray analysis, more replicates and a reduction in experimental complexity should be considered. It might also be necessary to apply less complex flow and process situations to be able to study

the shear response in more well-defined types of flow (Tanzeglock et al., 2009).

4. Conclusion

In this work, a Scale-Down Model of hydrodynamic stress conditions comparable to those found in a production scale bioreactor was developed to test a CHO production cell line for its robustness to hydrodynamic stress. Using 2L bioreactors operated at

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